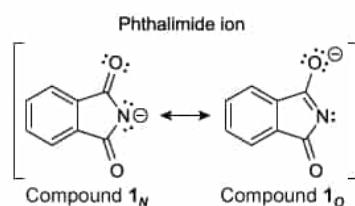
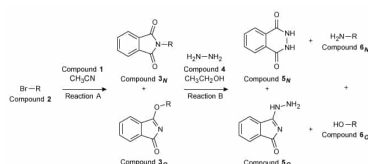


The phthalimide ion (Compound **1**) is the conjugate base of phthalimide, an aromatic heterocycle. Treatment with a hydroxide salt removes the acidic N–H proton from phthalimide and generates the resonance-stabilized phthalimide ion. Either of its resonance forms can act as a nucleophile; Compound **1_N** is the major resonance form and leads to *N*-alkylated products whereas Compound **1_O** is the minor resonance form and leads to *O*-alkylated products.



The nucleophilicity of the phthalimide ion plays a significant role in the Gabriel synthesis of primary amines (Scheme I). Treatment of a primary alkyl halide (Compound **2**) with Compound **1** promotes an S_N2 reaction (Reaction A) and generates a mixture of *N*- (Compound **3_N**) and *O*-alkylation (Compound **3_O**) products. Heating Compound **3** with an excess of hydrazine (Compound **4**) in ethanol facilitates the hydrazinolysis of Gabriel intermediates (Reaction B) and leads to the generation of poorly soluble hydrazine adducts (Compounds **5_N** and **5_O**) and the evolved primary amine (Compound **6_N**) or primary alcohol (Compound **6_O**).



Scheme I

Primary amines produced through the Gabriel synthesis are purified through a multistep procedure. Vacuum filtration removes insoluble material from the reaction mixture. After concentrating the reaction solvent under vacuum, the amine is extracted into an organic layer through liquid-liquid extraction. Following the concentration of extraction solvent under vacuum, distillation and/or normal-phase chromatography further separates the amine from other contaminants. Analysis of products is amenable to most methods of organic spectroscopy, including ¹H NMR spectroscopy, infrared spectroscopy, normal-phase high-performance liquid chromatography (NP-HPLC), and mass spectrometry. The Gabriel synthesis was performed using a model system (R = benzyl); relevant molecular data are shown in Table 1 below.

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Compound	R	Molecular weight (g mol ⁻¹)	Melting point (°C)	Boiling point (°C)	Solubility in ethanol
2	benzyl	171	–4	193	miscible
4	N/A	32	2	114	miscible
5_N	N/A	162	344	>300	very low
6_N	benzyl	107	10	185	miscible
6_O	benzyl	108	–15	205	miscible

After Reaction B is complete, its contents are cooled to room temperature (20 °C) and a colorless solid precipitate is formed. Based on the information in the passage, the major component of the precipitate is:

- ☐ A. [12%]
- ☒ B. [73%]
- ☐ C. [5%]
- ☐ D. [8%]

Correct
73% answered correctly

Explanation:

Identity of a precipitate in a hydrazinolysis reaction

Compound	R	Molecular weight (g/mol)	Melting point (°C)	Boiling point (°C)	Solubility in ethanol
2	benzyl	171	-4	193	miscible
4	H ₂ N	32	2	114	miscible
5_N	H ₂ N	162	344	>300	very low
6_N	benzyl	197	10	185	miscible
6_O	benzyl	198	-15	205	miscible

25 °C

Ethanol solution

- Compound **2**
- Compound **4**
- Compound **6_N**
- Compound **6_O**

Insoluble precipitate

- Compound **5_N** (major component)
- Compound **5_O** (minor component)

Compound **5_N** is the **least soluble** component of the reaction mixture and is the major component of the precipitate.

Compound **5_N**

O=C1NC(=O)c2ccccc21

The **solubility** of a substance is a quantitative measure of the maximum mass that can become dissolved in a specific volume of solvent. Solubility trends typically adhere to the premise of "like dissolves like," where solubility increases when the **polarity** of the solute is similar to the polarity of the solvent. Since a compound's polarity results from the presence and spatial arrangement of functional groups, solubility is often directly related to molecular structure and its **intermolecular forces** with other materials.

The solubility of **solid** solutes tends to increase with temperature. The heat provides additional kinetic energy to overcome the intermolecular forces between solid molecules and between solvent molecules. Consequently, cooling a solution can form a **precipitate** if the solubility of the solid solute decreases and the quantity of dissolved solute exceeds the new, lower solubility limit.

Reaction B is performed on a mixture of Compounds **3_N** and **3_O** using an excess of Compound **4** and heat. As a result, the solubility of all reaction components is greater throughout the heated reaction period. As the solution is cooled to room temperature, the solid precipitate is the component with the **lowest solubility**.

The passage states that **hydrazine adducts** (Compounds **5_N** and **5_O**) have poor solubility in ethanol. In contrast, the other reaction components (ethanol; Compounds **4**, **6_N**, and **6_O**) are liquids at 25 °C and are **miscible** in ethanol. Given that *N*-alkylation (Compound **3_N**) is the major route for Reaction A, Compound **5_N** (the structure shown in Choice B) is the major component of the precipitate following Reaction B.

(Choices A and C) Compounds **6_N** and **6_O** are *miscible* in the reaction solvent, ethanol. In addition, the melting point of both compounds is below 25 °C, and they will be *liquids* at this temperature.

(Choice D) Hydrazine (Compound **4**) has a melting point below the given conditions, and it is miscible in ethanol. As such, this choice is unlikely to be the solid precipitate.

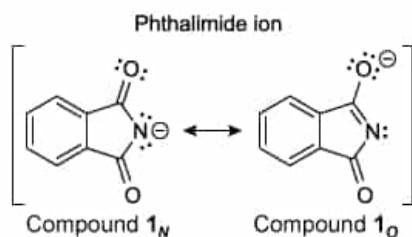
Educational objective:

The solubility of solid solutes increases with temperature and is influenced by the presence of functional groups. The identity of a precipitate can sometimes be elucidated through logical evaluation of the structures and physical properties of materials present in a mixture.

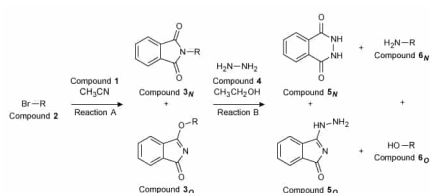
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Subject:
Organic Chemistry

QID: 404053
System:
Separation Techniques, Spectroscopy, and Analytical Methods

The phthalimide ion (Compound **1**) is the conjugate base of phthalimide, an aromatic heterocycle. Treatment with a hydroxide salt removes the acidic N–H proton from phthalimide and generates the resonance-stabilized phthalimide ion. Either of its resonance forms can act as a nucleophile; Compound **1_N** is the major resonance form and leads to *N*-alkylated products whereas Compound **1_O** is the minor resonance form and leads to *O*-alkylated products.



The nucleophilicity of the phthalimide ion plays a significant role in the Gabriel synthesis of primary amines (Scheme I). Treatment of a primary alkyl halide (Compound **2**) with Compound **1** promotes an S_N2 reaction (Reaction A) and generates a mixture of *N*- (Compound **3_N**) and *O*-alkylation (Compound **3_O**) products. Heating Compound **3** with an excess of hydrazine (Compound **4**) in ethanol facilitates the hydrazinolysis of Gabriel intermediates (Reaction B) and leads to the generation of poorly soluble hydrazine adducts (Compounds **5_N** and **5_O**) and the evolved primary amine (Compound **6_N**) or primary alcohol (Compound **6_O**).



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Primary amines produced through the Gabriel synthesis are purified through a multistep procedure. Vacuum filtration removes insoluble material from the reaction mixture. After concentrating the reaction solvent under vacuum, the amine is extracted into an organic layer through liquid-liquid extraction. Following the concentration of extraction solvent under vacuum, distillation and/or normal-phase chromatography further separates the amine from other contaminants. Analysis of products is amenable to most methods of organic spectroscopy, including ¹H NMR spectroscopy, infrared spectroscopy, normal-phase high-performance liquid chromatography (NP-HPLC), and mass spectrometry. The Gabriel synthesis was performed using a model system (R = benzyl); relevant molecular data are shown in Table 1 below.

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6_O	benzyl	108	–15	205	miscible

Water is added to the concentrated residue from Reaction B. Which liquid is the most likely to be effective in extracting the desired amine product?

- ☐ A. Methanol [6%]
☐ B. Ethanol [19%]
☐ C. *n*-Hexane [48%]
☒ D. Ethyl acetate [24%]

Correct

25% answered correctly

Explanation:

Determining the best solvent for an organic extraction



Solvent	Miscible with water?	Organic polarity	Solubility of Compound 6_N
methanol	yes	polar	high
ethanol	yes	polar	high
<i>n</i> -hexane	no	nonpolar	low
ethyl acetate	no	polar	high

- Organic liquid
- Immiscible with water
- Moderate polarity

Nc1ccccc1 + Other solutes + H₂O

Compound **6_N**

- Organic-soluble
- Moderate polarity

- Mixture
- Ideally are aqueous-soluble

- Aqueous
- Polar

Ethyl acetate has the preferred characteristics for the liquid-liquid extraction of Compound **6_N**.

A **liquid-liquid extraction** is a laboratory technique that facilitates the separation of materials based on differences in **solubility**. Liquid-liquid extraction requires liquids that are not soluble in one another (**immiscible**) and form two distinct layers when placed in a container. Extraction liquids typically have different polarities, and dissolved solute(s) partition between the two layers based on the principle of "**like dissolves like**."

A **separatory funnel** is used to perform liquid-liquid extractions. The liquid with the higher density will be the bottom layer in the funnel and can be drained into a different container. The liquid with the lower density remaining in the separatory funnel is then poured out the top of the separatory funnel. Liquid-liquid extractions are most effective when performed **iteratively**, where the layer requiring extraction is returned to the funnel and the opposing immiscible liquid is added to repeat the extraction process.

In this question, water is added to the residue of Reaction B. Water is polar and tends to dissolve polar solutes and ions. Consequently, the second liquid needs to be an **organic liquid that does not dissolve in water**. To effectively

extract the polar organic amine (Compound **6_N**), the **organic layer must also be moderately polar**. Ethyl acetate is the only liquid listed that meets the criteria of being both moderately polar and immiscible in water.

(Choices A and B) **Alcohols** with fewer than three carbon atoms per **hydroxyl group** are miscible in water. Neither methanol (1-carbon) nor ethanol (2-carbon) can be used for a liquid-liquid extraction with water as the second solvent, as a mixture of the alcohol and water *will not form* the two distinct layers required for separation.

(Choice C) Although *n*-hexane meets the criteria of being immiscible in water, it is a purely **nonpolar** organic solvent. It is **unlikely** that the desired polar amine (Compound **6_N**) would have high enough solubility in *n*-hexane for the process to be an effective extraction.

Educational objective:

Liquid-liquid extraction is the process of separating components in a mixture using the differential solubility of solutes in two immiscible liquids. The criteria for selection of appropriate liquid(s) can vary between experimental applications and must be evaluated on a case-by-case basis.

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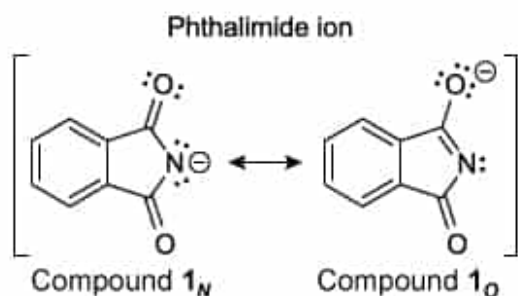
Subject:
Organic Chemistry

QID: 404054

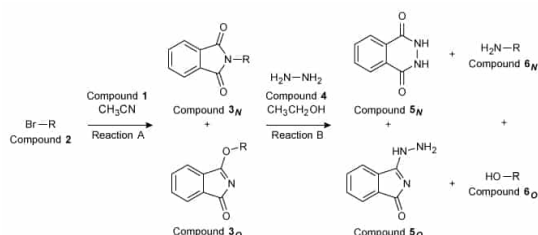
System:

Separation Techniques, Spectroscopy, and Analytical Methods

The phthalimide ion (Compound **1**) is the conjugate base of phthalimide, an aromatic heterocycle. Treatment with a hydroxide salt removes the acidic N–H proton from phthalimide and generates the resonance-stabilized phthalimide ion. Either of its resonance forms can act as a nucleophile; Compound **1_N** is the major resonance form and leads to *N*-alkylated products whereas Compound **1_O** is the minor resonance form and leads to *O*-alkylated products.



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Primary amines produced through the Gabriel synthesis are purified through a multistep procedure. Vacuum filtration removes insoluble material from the reaction mixture. After concentrating the reaction solvent under vacuum, the amine is extracted into an organic layer through liquid-liquid extraction. Following the concentration of extraction solvent under vacuum, distillation and/or normal-phase chromatography further separates the amine from other contaminants. Analysis of products is amenable to most methods of organic spectroscopy, including ¹H NMR spectroscopy, infrared spectroscopy, normal-phase high-performance liquid chromatography (NP-HPLC), and mass spectrometry. The Gabriel synthesis was performed using a model system (R = benzyl); relevant molecular data are shown in Table 1 below.

Table 1 Molecular Data for Compounds Used in the Model System for the Gabriel Synthesis

Compound	R	Molecular weight (g mol ⁻¹)	Melting point (°C)	Boiling point (°C)	Solubility in ethanol
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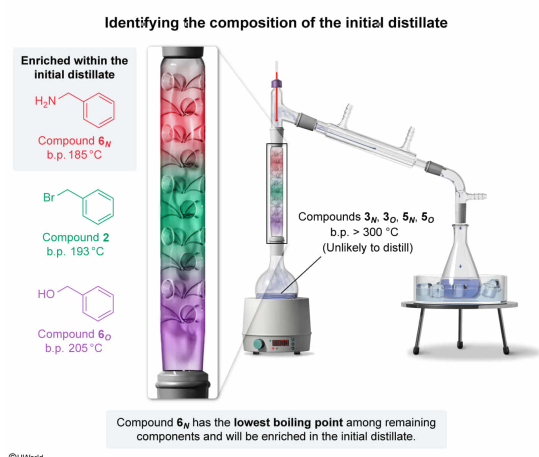
When the organic extract following Reaction B is subjected to fractional distillation, what can be inferred about the initial distillate?

- ✓ ☒ A. Compound **6_N** will be enriched in the initial distillate. [55%]
- ☐ B. Compound **2** will be enriched in the initial distillate. [12%]
- ☐ C. Compound **6_O** will be enriched in the initial distillate. [22%]
- ☐ D. Nothing can be inferred about the composition of the initial distillate. [10%]

Correct

54% answered correctly

Explanation:



Distillation is a technique used to separate compounds based on their boiling point. There are three common types of distillation: **simple**, **fractional**, and **vacuum**. The type of distillation appropriate for a mixture depends on the boiling points (b.p.) of the mixture components.

Fractional distillation works best for compounds that have boiling points that are less than 25 °C apart, and employs a fractionating column to provide additional surface area for the distillate vapors to iteratively condense and evaporate before collection in the receiving flask.

This question asks what can be inferred about the composition of the initial distillate following extraction of Reaction B. When the extraction solvent (ie, ethyl acetate, b.p. 77 °C) is evaporated, low-boiling reaction components (ie, Compound **4**, b.p. 114 °C) are likely to also be evaporated. Similarly, any residual amounts of phthalimide-derived compounds (ie, Compounds **3_N, 3_O, 5_N, 5_O**) in the extract are unlikely to distill, given their high boiling points (greater than 300 °C).

With these assumptions, Compounds **2** (b.p. 193 °C), **6_N** (b.p. 185 °C), and **6_O** (b.p. 205 °C) are most likely relevant to the distillation. Since all three compounds fall within a 25 °C range, fractional distillation will be the most effective distillation method. Of these, **Compound 6_N** has the **lowest boiling point**, and it can be inferred that Compound **6_N** will proceed through the fractional distillation the quickest and be **enriched in the initial distillate**.

(Choice B) Compound **2** has the *intermediate* boiling point (193 °C) among the three relevant compounds. Therefore, it can be inferred that Compound **2** would be enriched within *intermediate* fractions during the distillation.

(Choice C) Compound **6_O** has the *highest* boiling point (205 °C) among the three relevant compounds. Therefore, it can be inferred that Compound **6_O** would be enriched within either the *final* fractions of the distillation or the residue remaining in the distillation flask.

(Choice D) The passage includes boiling point data for most components in the model Gabriel amine synthesis, which allows inference about the composition of the initial distillate.

Educational objective:

Distillation is a technique used to separate a mixture of liquids based on a difference in boiling points. The initial distillate in a fractional distillation will be enriched in the lowest-boiling component that was placed in the distillation flask.

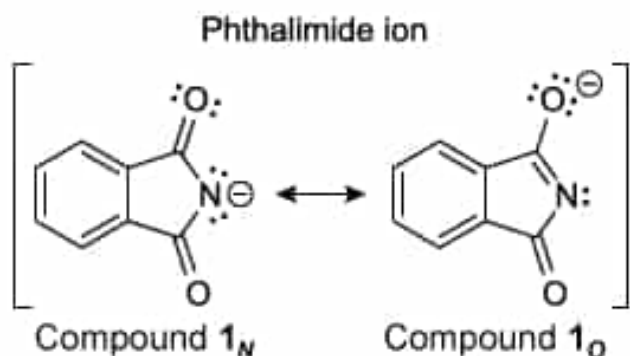
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Subject:
Organic Chemistry

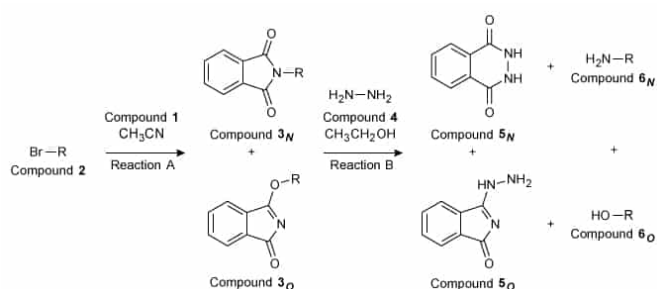
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System:
Separation Techniques, Spectroscopy, and Analytical Methods

The phthalimide ion (Compound **1**) is the conjugate base of phthalimide, an aromatic heterocycle. Treatment with a hydroxide salt removes the acidic N–H proton from phthalimide and generates the resonance-stabilized phthalimide ion. Either of its resonance forms can act as a nucleophile; Compound **1_N** is the major resonance form and leads to *N*-alkylated products whereas Compound **1_O** is the minor resonance form and leads to *O*-alkylated products.



The nucleophilicity of the phthalimide ion plays a significant role in the Gabriel synthesis of primary amines (Scheme I). Treatment of a primary alkyl halide (Compound **2**) with Compound **1** promotes an S_N2 reaction (Reaction A) and generates a mixture of *N*- (Compound **3_N**) and *O*-alkylation (Compound **3_O**) products. Heating Compound **3** with an excess of hydrazine (Compound **4**) in ethanol facilitates the hydrazinolysis of Gabriel intermediates (Reaction B) and leads to the generation of poorly soluble hydrazine adducts (Compounds **5_N** and **5_O**) and the evolved primary amine (Compound **6_N**) or primary alcohol (Compound **6_O**).



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Primary amines produced through the Gabriel synthesis are purified through a multistep procedure. Vacuum filtration removes insoluble material from the reaction mixture. After concentrating the reaction solvent under vacuum, the

amine is extracted into an organic layer through liquid-liquid extraction. Following the concentration of extraction solvent under vacuum, distillation and/or normal-phase chromatography further separates the amine from other contaminants. Analysis of products is amenable to most methods of organic spectroscopy, including ^1H NMR spectroscopy, infrared spectroscopy, normal-phase high-performance liquid chromatography (NP-HPLC), and mass spectrometry. The Gabriel synthesis was performed using a model system ($\text{R} = \text{benzyl}$); relevant molecular data are shown in Table 1 below.

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6_O	benzyl	108	-15	205	miscible

The organic extract from Reaction B is primarily composed of three materials: Compound **2**, Compound **6_N**, and Compound **6_O**. Which statement most accurately describes the relationship between the expected order of elution for these three compounds via NP-HPLC and fractional distillation?

- ☐ A. The order of elution is the same for NP-HPLC and fractional distillation [33%]
- ☐ B. The order of elution is the opposite for NP-HPLC and fractional distillation [25%]
- ☒ C. The order of elution is different for NP-HPLC and fractional distillation [39%]
- ☐ D. Nothing can be inferred about the relative order of elution for NP-HPLC and fractional distillation [2%]

Correct

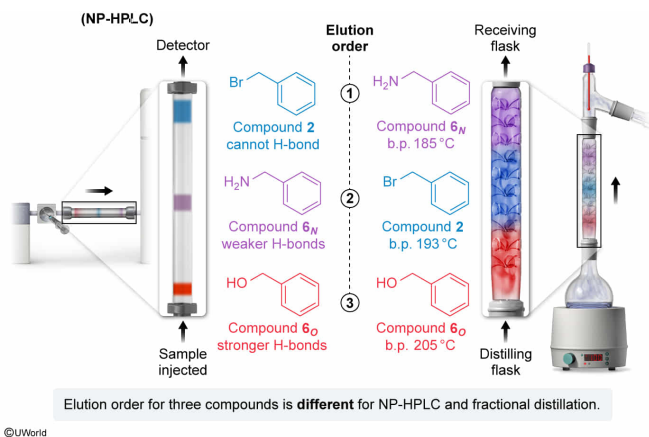
39% answered correctly

Explanation:

Comparing the elution order for separation techniques

Normal-phase HPLC

Fractional distillation



Normal-phase (NP) high-performance liquid chromatography (**HPLC**) separates materials based on affinity for the nonpolar mobile phase (eg, hexanes, ethyl acetate) and the polar stationary phase (eg, silica, alumina). Due to the principle of "**like dissolves like**," nonpolar compounds that more strongly interact with the nonpolar mobile phase travel faster through the column whereas compounds that **more strongly interact** with the polar stationary phase travel slower through the column. Therefore, in NP-HPLC, **nonpolar compounds elute faster** than more polar compounds.

Distillation is used to separate compounds based on their **boiling point** (b.p.), and compounds distill in order of **increasing boiling point**. **Fractional distillation** uses a **fractionating column** to provide additional surface area for vapors to iteratively condense and evaporate before collection.

Comparing Compounds **2**, **6_N**, and **6_O** and their interactions with the polar stationary phase in NP-HPLC:

- Compound **2** (an alkyl halide) is the most nonpolar as it contains a large, **polarizable** halogen atom and cannot **hydrogen bond**.
- Compounds **6_N** (an **amine**) and **6_O** (an **alcohol**) can both hydrogen bond.
- The hydrogen bonds for Compound **6_N** are weaker given a smaller **electronegativity** difference of the N–H bond.

Therefore, the order of elution via NP-HPLC is Compound **2** → Compound **6_N** → Compound **6_O**.

The separation of Compounds **2** (b.p. 193 °C), **6_N** (b.p. 185 °C), and **6_O** (b.p. 205 °C) using fractional distillation is based on differences in boiling point, with the lowest boiling point compound distilling (eluting) first. Therefore, the order

of elution via fractional distillation is Compound **6_N** → Compound **2** → Compound **6_O**.

The most accurate conclusion is that the **elution order for these compounds is different (but not opposite)** for NP-HPLC and fractional distillation.

(Choice A) The order of elution is *not* the same for these separation methods, as polarity and boiling point orders *do not* follow the same trends.

(Choice B) An opposite order of elution would be expected if the separation processes were reciprocal in nature (ie, normal-phase HPLC vs. [reverse-phase HPLC](#)). Fractional distillation and NP-HPLC use different criteria to determine elution order.

(Choice D) Sufficient information is provided in the passage to independently determine the elution order for compounds using NP-HPLC and fractional distillation. Comparison of methods allows conclusions to be drawn about relative elution order.

Educational objective:

NP-HPLC and fractional distillation are two methods that can be employed to separate mixtures. Different separation methods employ different chemical principles to facilitate separation, which can lead to differences in elution order.

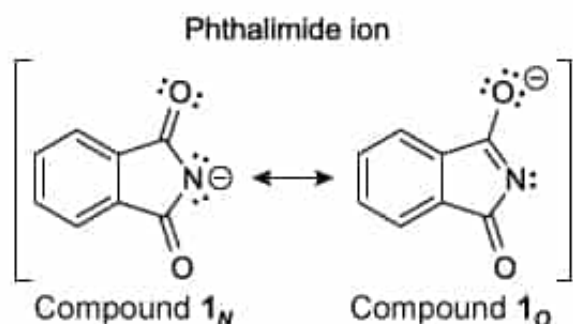
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Subject:
Organic Chemistry

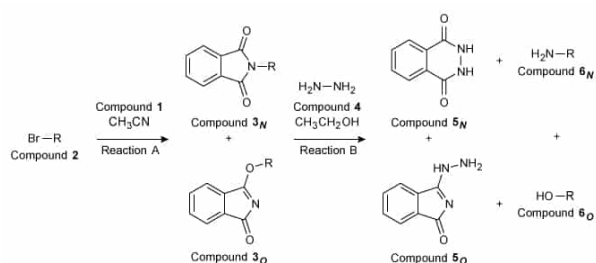
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System:
Separation Techniques, Spectroscopy, and
Analytical Methods

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Scheme I

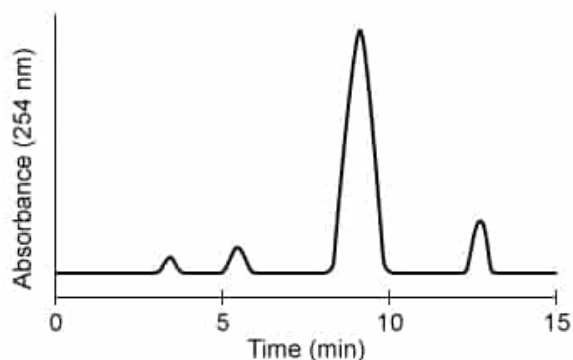
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Following completion of the Gabriel synthesis using the model system (R = benzyl), analysis of the organic extract from Reaction B using NP-HPLC generated the following chromatogram. Analysis of standards determined retention times for the following compounds: Compound **2** (3.8 min), **3_N** (5.2 min), **6_N** (9.2 min), and **6_O** (11.4 min). This experiment shows that:



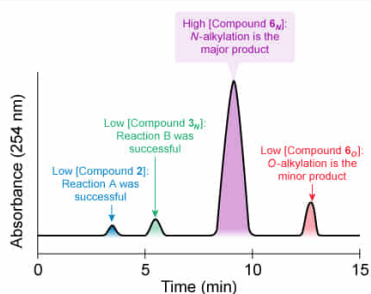
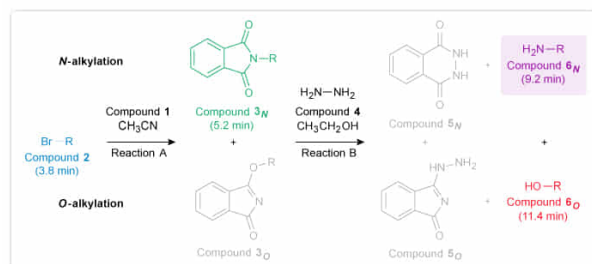
- ☐ A. only Reaction A was unsuccessful. [3%]
- ✓ ☒ B. the *N*-alkylation product was the major product. [87%]
- ☐ C. the *O*-alkylation product was the major product. [6%]
- ☐ D. only Reaction B was unsuccessful. [2%]

Correct

87% answered correctly

Explanation:

Drawing conclusions from an HPLC chromatogram



Normal-phase (NP) high-performance liquid chromatography (**HPLC**) separates materials based on affinity for the nonpolar mobile phase and the polar stationary phase. Due to the principle of "**like dissolves like**," nonpolar compounds that more strongly interact with the nonpolar mobile phase travel faster through the column whereas compounds that **more strongly interact** with the polar stationary phase travel slower through the column. In NP-HPLC, **nonpolar compounds have a shorter retention time**, and **polar compounds have a longer retention time**.

HPLC data is typically presented as a **chromatogram** that depicts quantities of analyte (y-axis) as a function of time (x-axis). HPLC systems use a detector to quantify the amount of material that passes the sensor. A common detector type measures the absorption of light in the ultraviolet-visible (UV-VIS) region of the **electromagnetic spectrum**. Compounds containing an **aromatic ring** tend to absorb strongly around a wavelength of **254 nm**. Applying a variation of **the Beer-Lambert law**, the **area under the curve (AUC)** for a peak is proportional to the quantity of material present.

The appearance of four peaks in the given chromatogram (recorded at 254 nm) suggests the presence of four aromatic analytes. Peaks at 3.8, 5.2, and 11.4 min have a small AUC, but the peak at 9.2 min has a much larger AUC. Based on the provided retention times, the data demonstrates that Compound **6_N** is the major component of the product sample. Given that Compound **6_N** is generated following **N-alkylation** of the phthalimide ion in Reaction A, it can be concluded that the **N-alkylation product was the major product of this experiment**.

(Choices A and D) If Reaction A or B was unsuccessful, a large amount of unreacted Compound **2** (retention time 3.8 min) or Compound **3_N** (retention time 5.2 min),

respectively, would be present in the chromatogram. However, the AUC for the peaks at 3.8 min and 5.2 min are *small*, suggesting that Reactions A and B were *successful*.

(Choice C) Compound **6_O** (retention time 11.4 min) is formed following O-alkylation of the phthalimide ion. The AUC for the peak at 11.4 min is *small*, suggesting that O-alkylation is a *minor* product.

Educational objective:

An HPLC chromatogram depicts the quantity of analyte as a function of retention time. In the context of an experimental procedure, analysis of chromatogram AUCs allows conclusions to be drawn about experimental outcomes.

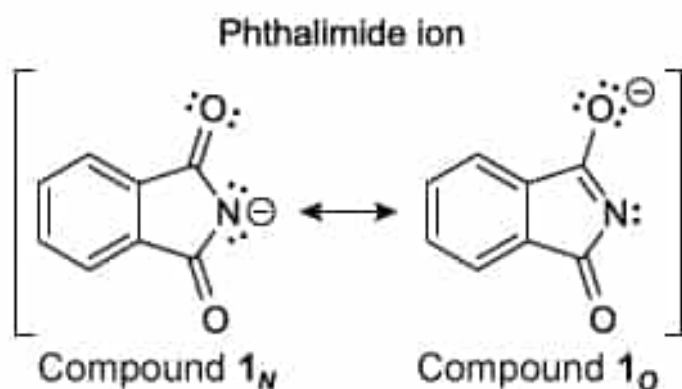
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Subject:
Organic Chemistry

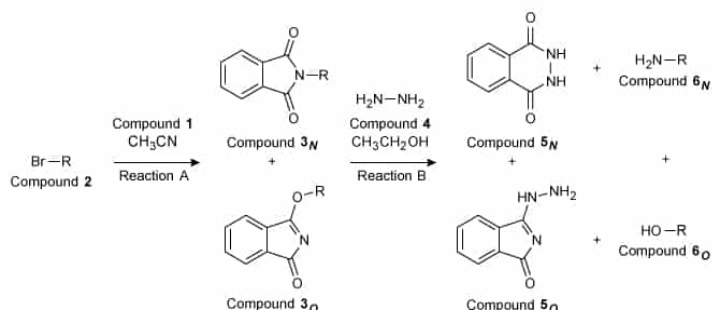
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System:
Separation Techniques, Spectroscopy, and Analytical
Methods

The phthalimide ion (Compound **1**) is the conjugate base of phthalimide, an aromatic heterocycle. Treatment with a hydroxide salt removes the acidic N–H proton from phthalimide and generates the resonance-stabilized phthalimide ion. Either of its resonance forms can act as a nucleophile; Compound **1_N** is the major resonance form and leads to *N*-alkylated products whereas Compound **1_O** is the minor resonance form and leads to *O*-alkylated products.



The nucleophilicity of the phthalimide ion plays a significant role in the Gabriel synthesis of primary amines (Scheme I). Treatment of a primary alkyl halide (Compound **2**) with Compound **1** promotes an S_N2 reaction (Reaction A) and generates a mixture of *N*- (Compound **3_N**) and *O*-alkylation (Compound **3_O**) products. Heating Compound **3** with an excess of hydrazine (Compound **4**) in ethanol facilitates the hydrazinolysis of Gabriel intermediates (Reaction B) and leads to the generation of poorly soluble hydrazine adducts (Compounds **5_N** and **5_O**) and the evolved primary amine (Compound **6_N**) or primary alcohol (Compound **6_O**).



Scheme I

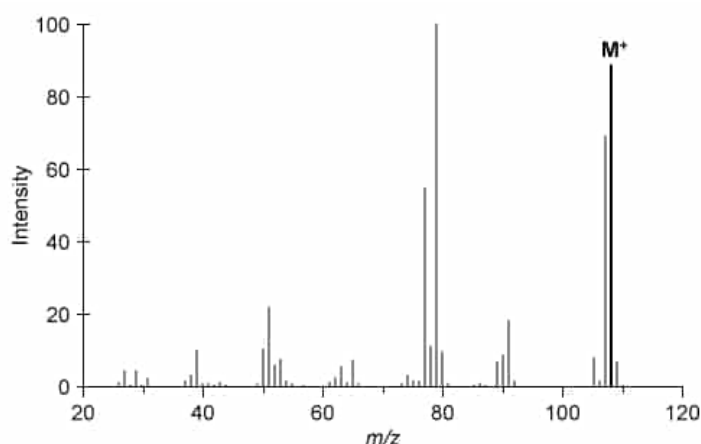
Primary amines produced through the Gabriel synthesis are purified through a multistep procedure. Vacuum filtration removes insoluble material from the reaction mixture. After concentrating the reaction solvent under vacuum, the amine is extracted into an organic layer through liquid-liquid extraction. Following the concentration of extraction solvent under vacuum, distillation and/or normal-phase chromatography further separates the amine from other contaminants.

Analysis of products is amenable to most methods of organic spectroscopy, including ^1H NMR spectroscopy, infrared spectroscopy, normal-phase high-performance liquid chromatography (NP-HPLC), and mass spectrometry. The Gabriel synthesis was performed using a model system ($\text{R} = \text{benzyl}$); relevant molecular data are shown in Table 1 below.

Table 1 Molecular Data for Compounds Used in the Model System for the Gabriel Synthesis

Compound	R	Molecular weight (g mol^{-1})	Melting point ($^{\circ}\text{C}$)	Boiling point ($^{\circ}\text{C}$)	Solubility in ethanol
2	benzyl	171	-4	193	miscible
4	N/A	32	2	114	miscible
5 _N	N/A	162	344	>300	very low
6 _N	benzyl	107	10	185	miscible
6 _O	benzyl	108	-15	205	miscible

The mass spectrum of a compound from Gabriel synthesis of the model system ($\text{R} = \text{benzyl}$) is shown below. Which of the following compounds is consistent with the mass spectrum data?

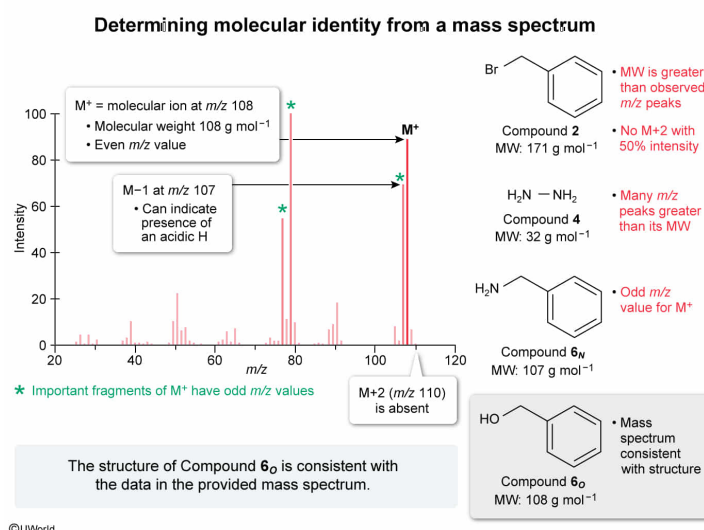


- ☐ A. Compound **2** [18%]
☐ B. Compound **4** [9%]
☐ C. Compound **6_N** [34%]
☒ D. Compound **6_O** [37%]

Correct

36% answered correctly

Explanation:



Mass spectrometry provides the ability to measure the mass of molecules and molecular fragments. An **electron-impact mass spectrometer** bombards a sample with electrons to remove an electron from a molecule (**ionization**). This creates the **molecular ion (M⁺)**, a radical cation with a **mass-to-charge (m/z) ratio** equal to the molecular weight of the molecule. The molecular ion can also break into a radical and a cation to produce smaller molecular fragments. Finally, a **mass spectrum** plot displaying the abundance of detected cations (y-axis)

versus m/z (x-axis) is generated.

The presence of functional groups leads to characteristic signatures in a mass spectrum:

- **Amides**, **amines**, and compounds that contain an odd number of nitrogen atoms have a molecular ion with an odd m/z and fragments with an even m/z .
- Some **alcohols** are prone to **dehydration**, where the molecular ion is apparent 18 m/z below the expected value.
- Chlorinated and brominated compounds have peaks determined by their natural **isotopic abundance**.

The mass spectrum in this question has a M^+ at m/z 108, an even value that matches the molecular weight of Compound **6_O**. Although Compound **6_O** is an alcohol, its structure precludes the ability to undergo dehydration as it lacks the necessary proton at the carbon adjacent to the carbon bearing the alcohol functional group. The peaks at m/z 77 and m/z 91 are fragments that suggest the presence of a **benzyl group**. Together, the mass spectrum is consistent with the structure of Compound **6_O**.

(Choice A) As an alkyl bromide, the M^+ for Compound **2** would have M (^{79}Br , m/z 170) and $M+2$ (^{81}Br , m/z 172) peaks in approximately a 1:1 ratio. A bromine isotope signature is *not* present in the data, and the molecular ion occurs at a lower m/z .

(Choice B) Compound **4** has a low molecular weight (32 g mol^{-1}) that would have its M^+ at the same low m/z value. This mass spectrum has many higher m/z peaks.

(Choice C) As an amine, the M^+ for Compound **6_N** appears at m/z 107 (an *odd* value) with *even* m/z fragments. This spectrum has an *even* M^+ with *odd* m/z fragments.

Educational objective:

Mass spectrometry can be used to determine the identity of a compound. Analysis of the molecular ion and any isotope signatures can be used to support conclusions about functional groups that may be present in the sample.

Time spent:0 sec

Subject:
Organic Chemistry

QID:404058

System:
Separation Techniques, Spectroscopy, and
Analytical Methods